

Optimal Chemo Scheduling

First, observe from the code defining a_2 that a_2 is essentially 1 (in fact slightly less than 1 by an astronomically tiny amount). In fact, as $n \rightarrow \infty$ one shows

$$a_2 = 2 \left(\sum_{k=3}^{\infty} (1/2)^k - \sum_{k=3}^{\infty} 72^{-10^{23}k} \right) + \frac{1}{2} \approx 1.$$

Thus $a_2 < 1 = a_1$.

The optimization is

$$\min_{x_1, x_2 \geq 0} x_1 + x_2 \quad \text{s.t.} \quad x_1 + a_2 x_2 = 1.$$

Since $a_1 = 1$ is the larger rate, the shortest time is achieved by using only the first therapy. In other words, set $x_2 = 0$ (any $x_2 > 0$ would require extra time because $a_2 < 1$). Then the constraint $x_1 + a_2 x_2 = 1$ gives $x_1 = 1$.

Hence the optimal schedule (to two decimal places) is

$$x_1 \approx 1.00, \quad x_2 \approx 0.00.$$

This makes total dosage $1 \cdot x_1 + a_2 \cdot x_2 = 1 \cdot 1 + a_2 \cdot 0 = 1$ and minimizes the time $x_1 + x_2 = 1$.
